BIM Data and Sustainable Accessibility

Figure 1: BIM of the Berlage Bridge in Amsterdam (Scan2BIM)

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Involved Organizations: Engineering Department of the Municipality of Amsterdam, National Archives of the Netherlands, The Hague University of Applied Sciences

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Summary

This advisory report describes the findings of a pilot exploring the sustainable accessibility of BIM data, conducted by the Engineering Department of the Municipality of Amsterdam, the National Archives of the Netherlands, and The Hague University of Applied Sciences. The pilot aimed to answer the following research questions:

- To what extent is the Industry Foundation Classes (IFC) exchange standard suitable for archiving and preserving BIM data (sustainability)?
- To what extent are the viewers currently available on the market suitable for making BIM data accessible (accessibility)?

The advisory report elaborates on the following findings:

1. The archival perspective is missing from existing definitions of BIM data. Therefore, the pilot participants developed a new definition that describes BIM in a way that better meets the needs of the archival field.
2. The pilot explored various existing methods for making BIM data accessible via IFC viewers, specifically focusing on data protection and security. This report describes these findings and the challenges that arise.
3. During this pilot, various viewers were tested, including both free and commercial viewers. Open-source viewers often meet the criteria and needs of the archival sector but not those of the construction sector.
4. The flexibility with which the IFC standard can be interpreted leads to inconsistent data quality and interpretability. This poses challenges for the long-term accessibility of BIM data.

These findings led to the following recommendations:

1. Improve collaboration between the construction and archival sectors. Raise awareness within the construction sector about archival concepts such as the DUTO[1] principles or scan and Archiving by Design. By influencing the further development of the IFC standard, we can work together on solutions for the sustainable accessibility of the standard.
2. More collaboration is needed between the various future managers of BIM data, including municipalities, the national government, water boards, and provinces.
3. The project group extensively discussed issues surrounding free vs. paid software, viewing vs. editing and analyzing, and general public vs. experts. These issues remain unresolved. More research is needed. In the short term, the project group recommends focusing on implementing free viewers in our digital infrastructure.
4. Considering the two designated communities of the National Archives' e-depot (the general public and government record creators), the project group prefers free viewers with sufficient functionality to make BIM data available.
5. More knowledge sharing is needed about the management of BIM data between archival institutions, including those abroad.
6. Guidelines should be created for the archival management of IFC and BIM data to improve overall data quality.
7. The archival sector should seriously consider accepting IFC as a formal standard for the archival management of BIM data.

8. Conduct user research among the main target groups of BIM data.

[1)] *A DUTO scan is a requirements analysis in which user wishes for sustainable accessibility (DUTO) of information are translated into concrete recommendations for the organization. These recommendations initially apply within the scope defined for the DUTO scan, such as a work process. However, they could also apply to the entire organization.*

Executive Summary

This report describes the findings of a pilot study conducted by the Engineering Department of the Municipality of Amsterdam, the National Archives of the Netherlands, and The Hague University of Applied Sciences. Firstly, this pilot studied the Industry Foundation Classes (IFC) exchange standard and how it applies to archival purposes and the preservation of BIM/ As-Built information. Secondly, the pilot study explored whether the available viewers of today are suitable for providing access to BIM data in an archival context.

Several important findings from the pilot study are described in this report:

1. A new (additional) definition was created to describe BIM in a way that better suits the needs of the archival field.
2. Members of this pilot group explored several different methods that already exist to arrange access to BIM data through IFC viewers, specifically related to the themes of data protection and safety concerns. This report also describes the challenges that play a role.
3. Several freely available and commercial viewers were tested during this pilot study. Open-source viewers regularly meet the criteria and needs of the archival sector, but they lack the support and functionalities of the commercial (and costly) software the BIM sector prefers.
4. The flexibility with which the IFC standard can be interpreted and filled in leads to inconsistencies in the quality and interpretability of their data. This brings challenges with it for the long-term accessibility of BIM data.

These findings led the pilot group to recommend the following:

1. Improve collaboration between the construction and archival sectors. Create more awareness within the construction sector about the principles of DUTO and Archiving by Design.
2. The study group recommends more collaboration between the future stewards of BIM data. This includes government bodies such as municipalities, the national government, and provinces.
3. The study group debated for a long time on the questions of free-of-charge vs. paid software, viewing vs. processing and analyzing, and general public vs. experts. We need more research on these questions. For the short term, the study group recommends that the archival sector prioritizes the use of free-of-charge viewers.
4. From the perspective of the two designated communities of our e-depot (the general public and the creators of government archives), the study group prefers to offer free viewers with enough features to properly view BIM data, instead of offering whole BIM packages.
5. Exchange knowledge on the preservation of BIM data between archival institutions, including archival organizations in other countries.
6. Create archival guidelines for the usage of IFC and BIM data to improve overall data quality.
7. Seriously consider adopting IFC as a formal standard for archiving BIM data.
8. Conduct a user investigation into the main user groups of BIM data.

Index

Introduction

Before you lies the advisory report "BIM data and Sustainable Accessibility." This report is the result of a pilot organized by the Engineering Department of the Municipality of Amsterdam, the National Archives of the Netherlands, and The Hague University of Applied Sciences.

The National Archives participates in this pilot, led by the Engineering Department, with the following objectives:

- To explore the extent to which the Industry Foundation Classes (IFC) exchange standard is suitable for archiving and preserving BIM/As-Built information (sustainability).
- To explore the extent to which currently available viewers are suitable for making BIM data accessible (accessibility).

Due to differences in interpretations and interests, the pilot focuses on the interfaces between the construction sector and the archival field. The research findings are highlighted in this report, providing direction for follow-up research and shared solutions to promote the sustainable accessibility of BIM data for both traditional users of the archival sector and the construction sector.

Context and Motivation

The archival world is preparing for the provision and preservation of complex digital files. With the introduction of the new Archives Act, the current transfer period of 20 years will be shortened to 10 years after creation. The National Archives expects an increase in the transfer of digital-born archives and complex materials as a result.

BIM (Building Information Modelling) is widely used in the spatial sector, which includes both construction and geospatial sectors. This pilot is currently limited to the construction sector. Building information models are created in specific, primarily closed design software packages, but they are often exchanged and stored in an international open standard such as IFC. IFC is hosted by BuildingSMART, a global consortium of software developers, construction companies, and governments that collaboratively develop and use this format.

The construction sector has used BIM software since the 1980s, and in recent years, governments have also adopted building information models. A BIM contains performance requirements, (3D) drawings, material usage, construction methods, techniques, and measurements, making it a permanent part of the construction archive.

IFC is a crucial open standard for exchanging model information and is widely used in the spatial sector. However, it is essential that the standard also accounts for sustainable accessibility. Adjustments may be necessary in future versions, which aligns with BuildingSMART's iterative development approach.

The Environmental Act (Omgevingswet), effective January 1, 2024, has further emphasized this issue. Under this act, it is possible to submit an environmental permit application (building permit) using the IFC format. However, older building permits already in archives may not be sustainably accessible. In 2023, the Municipality of Amsterdam proposed a joint investigation with the National

Archives to explore the sustainable accessibility of IFC and assess the "IFC readiness" of the archival sector. This proposal led to this pilot.

Definition and Scope

Multiple definitions of BIM exist within the construction sector. Analysis during the pilot revealed that these definitions were insufficiently useful from the archival perspective. Therefore, participants developed a new definition aligned with the needs of the archival field:

"BIM is data linked to a digital representation of building elements, where the representation information remains connected to that data."

Note: This definition is expected to evolve in the future.

Considerations Regarding Existing BIM Definitions

To meaningfully archive information, it is crucial to understand how its structure contributes to its meaning. Most building information is developed as a relational database with an integrated graphical component, managed by software to represent traditional construction drawings. The Construction Digitization Council (BDR in The Netherlands) has issued a three-part definition as part of its "Knowledge Cards," widely used by BIM practitioners:

1. Building Information Model (BIM): A digital representation of how a building is designed, realized, and/or constructed.
2. Building Information Modelling: Emphasizes the process of collaborating in construction projects using digital models. Related concepts include integral design, concurrent engineering, lean planning, and digital information sharing.
3. Building Information Management: Focuses on the information itself—its structure, management, and reuse throughout the building's lifecycle.

The BDR considers all three meanings equally relevant. In this pilot, the first definition (Building Information Model) was primarily used, with "BIM data" serving as an overarching term.

Interfaces with Archiving

- The first two definitions relate to information relevant to a building's creation, emphasizing collaboration and digital representation.
- The third definition aligns with records management (archival management) but excludes archiving after a building's lifecycle, a critical aspect for the archival field.

During the pilot, it became clear that a BIM does not consist of a single integral model. Its meaning often derives from external sources that provide context for interpreting data. A sustainable link to external sources is essential, as it is impractical to include all referenced data in the archived dataset. This underscores the importance of open standards with robust management, such as IFC, for effective archiving.

Practical Example: BIM-Basis ILS (The Netherlands)

The BIM-Basic Information Delivery Specification (ILS), published in 2020, is a widely used standard in the Netherlands for defining the minimum information a BIM must contain and how it should be structured. This standard references IFC open standards for BIM communication (ISO 16739).

The Importance of "Archiving by Design"

"Archiving by Design" is a concept promoted by the National Archive in The Netherlands, which asserts that archiving begins at the moment information is created. Responsibility for archiving lies with the record creator (in this case, the construction sector). Archival repositories only become responsible for sustainable and accessible management after transfer.

In the construction sector, "Archiving by Design" is less familiar. The traditional assumption that archiving begins at the archival repository sometimes surfaces unconsciously. This pilot highlights the need to distinguish between "archiving" and "management by an archival repository."

Methodology

This pilot is an exploratory study with two key objectives:

- To explore whether the IFC exchange standard is suitable for archiving and preserving BIM/As-Built information (sustainability).
- To assess whether current market viewers are suitable for making BIM data accessible (accessibility).

Timeline and Team

- Duration: February 15 to August 27, 2024.
- Lead Researcher: Marin Rappard (Preservation Team, National Archives).
- Support: Jacob Takema, Remco van Veenendaal.
- Project Leader: Lauren Romijn (Relationship Management Team).
- Core Members: Theo Kremer, Ruud van der Meer (Municipality of Amsterdam), Ekko Nap (The Hague University of Applied Sciences).

Collaboration and Knowledge Sharing

Colleagues from the Knowledge & Advice, Collection, and Service Departments of the National Archives joined monthly for alignment. A public college morning on March 19, 2024, at The Hague University of Applied Sciences featured presentations on:

- 20 years of BIM
- Current use of BIM and IFC
- Linking IFC to metadata

These sessions revealed that BIM and IFC are far more complex than initially assumed, necessitating a test session for IFC viewers (June 4, 2024) and further source research.

Report Development

All project members contributed to the advisory report, which was finalized in collaboration with the National Archives' Communication Team.

Results

The following pages describe the results of the exploration of various interfaces that influence the sustainable accessibility of the IFC format.

1. The challenge regarding transparency and making a BIM available

The archival sector is familiar with incorporating non-public information. However, this usually concerns files or 'whole' documents. With a BIM, it may be the first time that within an information object, one part of the object requires a different security level than another part of the object. This is mainly due to safety requirements. For example, it may be that 'the public' is allowed to see where a door is located, but not which lock is used. This chapter elaborates on this topic, with examples of how security levels within the IFC format can be adjusted.

2. The flexibility of the IFC standard

IFC is an open standard. Open standards are used to enable the exchange of information created in proprietary software. The government and archival sectors also strongly encourage their use. Exchange between organizations is often necessary and essential for transferring information to an archive. However, the archive format usually does not have access to the software in which the information was created. An open standard makes it possible to open and display files. This chapter on the malleability of the IFC standard outlines some caveats of this open standard.

3. Providing accessibility with IFC viewers

To provide access to digital archives, viewers are needed. For the most standard formats (such as Office formats or PDFs), these viewers are integrated into the NA infrastructure, and users can view these formats seamlessly on the website or on-site. However, for many file formats, viewers are not yet available in the infrastructure. Therefore, one chapter also focuses on which viewers exist and their potential to serve as recommendations. This exploration provides insight into the advantages and disadvantages of the IFC standard for archiving BIM information.

4. Accountability and transparency about the construction process

The goals and use of BIM information differ between the construction and archival sectors. Broadly speaking, the construction sector focuses as much as possible on ease of use, efficiency, and collaboration. Monitoring the documentation of the process, context, and decision-making often takes a backseat. For archiving, however, this can be crucial information needed for accountability and transparency about the construction process, especially in the event of an incident in the built environment. This chapter discusses a supporting format (BIM Collaboration Format) that can give shape to this. It also outlines the current state of affairs regarding communication and documentation.

Based on these results, the following will be addressed:

- Recommendations to improve the sustainable accessibility of IFC;
- Advice on what research is needed for a follow-up trajectory;
- And what opportunities exist for archiving by design in the construction sector and standardization organizations such as BuildingSMART.

1. The challenge regarding transparency and making a BIM available

During the pilot, it became clear that transparency and accessibility would be a challenge. Various users want to view a BIM. However, not everyone has the right to the same level of access.

For example, a BIM is used by specialists who perform maintenance work. Citizens view the same BIM out of interest in architecture. When it comes to the BIM of a bunker from the Ministry of Defense, it is clear that these users should not all have the same level of access. But how do you organize this in a good way? In the pilot, the use of viewers was considered to provide a solution for this.

Impact of the New Archives Act in The Netherlands

The new Archives Act further sharpens this access issue. When this law becomes active, it will also affect the availability of BIM data. According to the current Archives Act, information is transferred to an archival repository 20 years after the closure of a file. Under the new act, this period is reduced to 10 years after creation. The entire access methodology must be adapted accordingly. The principle of transfer is that the information is exclusively held by the archival repository. This means that the archival repository oversees access to this information. This leads to complex fundamental questions: What happens when maintenance specialists can only find the necessary BIM data at the archival repository to perform their tasks? What happens when the fire department, after a disaster, immediately needs essential BIM data about a building on the verge of collapse? A new urgency arises around this fundamental question: Who gets access to what information, and how is this regulated?

Methods for Shielding Data

When weighing the requirements that the National Archives sets for methods to disclose BIM data using IFC, the issue of security is unavoidable. Securing disclosed data is not a new issue; it is an inherent part of working with BIM. Most Common Data Environment (CDE) applications have methods to manage data access:

- By making data (partially) unreadable through encryption.
- By replacing or omitting data.
- By dividing data.

A CDE is a shared digital space where all project information and documentation of a construction project is stored, managed, and exchanged. It serves as a central hub for collaboration, where team members have real-time access to the most current data and can work together on design and construction processes. Sometimes this is also called a Common Design Environment, as monitoring data is not managed in these systems.

In the context of this pilot, the key question is whether IFC also has a structure that can regulate access to data. A second relevant question is whether this issue has been considered before in other municipalities or other countries. This chapter focuses on the relationship between security, accessibility, and building data standards.

Method 1: Access Control

The IFC standard includes the IfcPermit(ACCESS) Class, which can record building data which individuals have specific access to the physical building. However, this concept cannot be related to access to the building's data. For this, a combination of one of the following methods is required. The expectation is that existing modules for displaying a BIM according to the IFC standard, in combination with this Class (IfcPermitACCESS), are not yet fully utilized. This expectation arises from the fact that the IFC standard administrator also issues certificates for applications that only partially implement the IFC schema.

Method 2: Making Data Unreadable

Access to data is restricted by encrypting, replacing, or omitting parts of the data. It is crucial that a viewer can still display the data structure later. It is possible to encrypt an object as a whole. However, due to the relationships an object has with other objects, the entire data set may become unreadable. This is a risk. There is an opportunity to configure a viewer so that an error in displaying one object does not immediately make reading other data impossible. For example, by showing a warningsign about the inability to display a related object. This keeps the viewer usable without requiring all data to be readable. To facilitate this, a recognizable marker can be placed in the inaccessible data, indicating whether a warning should be shown to a specific user. Of course, a specific value within an object's structure can also be made unreadable.

From an archival perspective, the above methods are not desirable. The National Archives wants information to comply with the DUTO characteristics: findable, available, readable, interpretable, reliable, and future-proof.

Method 3: Dividing Data

In the construction world, it is common for building data to come from multiple authors and to be shared for different purposes. Applications and modules for displaying building data are generally equipped to show data from different sources, known as aspect models. It makes sense to implement a security layer that restricts access to part of that data. This introduces a certain inefficiency. Each of these aspect models is a self-contained entity, with all the necessary parts of the data schema. What is also challenging in this context is establishing relationships between the different aspect models. For example, the IFC standard is not yet equipped to record relationships between parts of different models, although developments are underway towards this topic with IFC5 (or IFCX). Given the current situation, a viewer would not be able to report that an object with an accessible security level for a user also has related information with a different/higher security level.

Practical Example: NEN-EN-ISO 19650-5:2020

The sub-standard NEN-EN-ISO 19650-5:2020 addresses the "security-oriented approach to information management." This sub-standard is part of NEN-EN-ISO 19650, which pertains to building information management. NEN-EN-ISO 19650-5:2020 consists of five parts, with the last part focusing on security aspects. This part does not address implementation but does cover the mindset and culture needed to identify and mitigate vulnerabilities in an organization's data security. This ISO standard can serve as a first bridge between the archival field and the construction sector on the topic of security. The National Archives can use its own guidelines and standards for further supplementation. The standard NEN-EN-ISO 19650-5:2020 could provide starting points for setting up a viewer.

2. The Flexibility of the IFC Standard

To determine the sustainable accessibility of the IFC standard, an exploration of its structure, content, and possibilities is necessary. One challenge that arises is that there are many ways to implement the IFC standard. Additionally, different users have different requirements for the same BIM.

Developers are generally not focused on sustainable accessibility. Embedding a format with an open standard as an exchange format into an application is not the core business of a software company, not even for open-source design applications. The core business is developing new design possibilities, such as parametric design and the implementation of AI. The better the design tool, the more users switch to the product.

The IFC development (e.g., from IFC 1x2 to IFC 4x1 to IFCX) are closely followed by software companies. However, implementation always lags behind product development. This results in (open-source or paid) viewers showing different results. Sometimes objects are displayed differently, or the data is interpreted differently. Therefore, it is necessary to make agreements about this. A simple example is the length of a lamppost. Depending on whether a lamppost is still at the supplier's warehouse or has already been installed, the length may differ, but both lengths are correct. See the chapter on viewers for further elaboration.

Another example of the flexibility of the IFC standard is that IFC data can be encoded in various formats, such as XML, JSON, STEP, Turtle, RDF, OWL, HDF, IFC2Collada, or SQLite (see IFC formats). These formats can all be sent via web services and imported or exported. They can all be stored in files or managed in centralized or linked databases. It is up to the users themselves to decide what they want to share from their design tools via IFC. This can pose a risk to sustainable accessibility.

3. Providing Accessibility with IFC Viewers

In exploring the accessibility of the IFC standard, it is also important to look at the applications that support and display the standard. This chapter covers:

- The considerations involved.
- The user groups.
- The opportunities and challenges for the IFC format to provide access to BIM data.

- A comparison between commercial software (typically used in the construction sector) and open-source software (typically used in the archival sector).

Although developments around BIM, GIS, and the Dutch "Information Model Landuse Planning" (IMRO) information are monitored, archival services can currently often only bit-preserve (keeping the 1-s and 0-s intact and readable) this type of information. Sufficient knowledge and resources for functional preservation (keeping the information sustainably accessible) and display are lacking.

For providing and querying information, the preference in the archival sector often goes to a server-side solution. This makes the threshold for access as low as possible for the various user groups with different levels of knowledge. However, for increasingly complex digital-born files from different domains, this is not an easy task.

Before a server-side solution is implemented, the client-side is temporarily a possible solution. Specifically, this means that the archival service advises on viewers to use that can display the information object as correctly and authentically as possible. This chapter describes the two solutions in relation to each other, with concrete examples and suggestions.

Advantages of IFC Files

A major challenge for functional preservation is the fact that many sectors, including the construction sector, use a wide variety of software packages that produce output in their own file formats. These file formats are often not well interchangeable with organizations that use different software or even a different version of the same software. This was one of the main reasons for developing the IFC standard. Exchange between organizations is necessary during the construction process.

In the archival sector, there is also a strong push to use open standards such as IFC. Exchange between the record creator and the archival service is logically an essential part of the transfer process. But it can also play a role in the further management process. For example, because archival services usually do not have access to the required (expensive) software to open or display native files. This is also an advantage of the IFC standard: the number of applications that offer support is growing significantly and is not tied to a single vendor.

The way information is recorded in an IFC file is described so openly that the threshold is low for programmers to create something to display data. During the viewer test, a project member showed a viewer that was developed by a colleague in about half a day. This is a major advantage for accessibility.

Additionally, the IFC standard sets clear requirements for the structure of the file. There is also a strong movement towards increasing standardization within the standard, such as in the use and consistent application of naming conventions.

Note: IFC Means (Acceptable?) Data Loss

This chapter focuses on viewers for the IFC-SPF format. In the viewer tests, files with the .ifc extension, or STEP Physical Files, were mainly used in the project.

Figure 2: The IFC-formats (by buildingSMART)

Format	Extension	MIME Type	Text	Indexed	Size	Summary
Official						
STEP Physical File (SPF)	.ifc	application/x-step	Yes	No	100%	STEP Physical Format (SPF or IFC-SPF) is the most widely used format for IFC in practice, which is the most compact of the formats listed that can be read as text. IFC-SPF is based on the ISO standard for clear text representation of EXPRESS data models ISO 10303-21
Extensible Markup Language (XML)	.ifcXML	application/xml	Yes	No	113%	Extensible Markup Language (XML) provides enhanced readability and benefits from a broad range of software tools. ifcXML is based on the ISO standard for representation of STEP data in XML format ISO 10303-28
ZIP	.ifcZIP	application/zip	No	No	17%	IFC data may be embedded within a ZIP file. The embedded data may be encoded as either SPF or XML, where the resulting size is typically comparable.
Terse RDF Triple Language (Turtle)	.ttl based on ifcOWL	text/turtle	Yes	No	1372%	More info on: ifcOWL
Resource Description Framework (RDF/XML)	.rdf based on ifcOWL	application/rdf+xml	Yes	No	816%	More info on: ifcOWL
Provisional/ Candidate						
JavaScript Object Notation (JSON)	.json	application/json	Yes	No	148%	JSON provides enhanced readability and benefits from a broad range of software tools.
Hierarchical Data Format (HDF)	.hdf	application/x-hdf	No	Yes	n/a	HDF5 may store IFC data within hierarchical database, which provides high performance access to engineering data. HDF is based on the ISO standard for STEP data representation ISO 10303-26
Experimental/ Unsupported						
SQLite	.sqlite	application/x-sqlite3	No	Yes	n/a	SQLite may store IFC data within a relational database, which provides indexed access to data within large models and benefits from a broad range of software tools.

An important note here is that these are not the native formats in which the building information was likely created. This means that, in practice, a certain degree of data loss occurs when converting to IFC. Choosing IFC as an archiving standard therefore requires a level of comfort with (some degree of) data loss. Certain functionalities that exist in the original software may be lost. However, the development of IFC is not yet complete. It is possible that data loss will decrease over time, as seen with the release of new versions of the standard. Theo Kremer and Ruud van der Meer emphasized that in Amsterdam, the strategy is to preserve both the native formats and an IFC file for archiving, as different manifestations. Given the scope of this pilot, this is not considered here.

Challenges for the Accessibility of IFC

However, the IFC format also has challenges that affect accessibility. The flexibility of the standard allows for a great deal of freedom in its application. Organizations can, for example, use their own properties and property sets. Without a documented object library, these are not understandable to the layperson or even the expert.

From the test session, it emerged that one software company created its own names for properties. This freedom exists and can pose a risk for archiving. The so-called Object Type Library (OTL) is therefore necessary representational information that is not stored within the IFC standard but in external documentation. This must always be provided. For these cases, merely displaying the model with metadata is probably not sufficient. If this is not handled consciously, virtually useless IFC models may be received that can be opened but whose metadata is not understandable. Solutions are being developed for this, such as enforcing the inclusion of ILS and OTL, and the development of the buildingSMART Data Dictionary (bSDD). See also "Interaction of IFC and Supporting Formats."

Another factor that makes advising on viewers difficult is that they can be certified for parts of the IFC standard rather than covering the entire scope of the standard. A possible consequence is that when trying to view an IFC with functionalities that fall outside the certified part, the display is not entirely authentic. Authenticity is, of course, important. Experts indicate that there are almost no viewers, commercial or open-source, that are certified for every functionality of the IFC standard. The estimate is that the most used functions are supported by most viewers. However, research is needed to determine the extent of cases that fall outside this scope.

It is therefore extra important to include in the requirements that the data is clearly defined. Validation can play a role in this. With the advice or requirement to validate that a model covers a certain part of the IFC standard before transfer.

Interaction of IFC and Supporting Formats

From the test session, it emerged that parts of the construction process (such as decision-making/communication) that may be important for long-term preservation and the ability to provide accountability are not stored in the IFC file but in supporting formats. For communication and decision-making, for example, the BIM Collaboration Format (BCF) is important. Special attention is given to this format in the next chapter. Via persistent identifiers (GUIDs), BCF communication is linked to the specific parts in the IFC model that the communication concerns. The ability to view this communication can be important, especially with a view to transparency and accountability of decisions made.

To correctly interpret all metadata, information is needed about the Object Type Library used. The Information Delivery Specification (ILS) is intended to make a BIM more "exchangeable, structured, unambiguous, correct, complete, and reusable." Recently, IDS was added to the arsenal to further improve data quality.

There is therefore an important "package" of information that must be delivered with IFC. It is therefore important to focus on providing the necessary documentation. In various reports and papers, the advice is to create and deliver a document at the end of a project that records the process, governance, and context to provide "further access" to users. This serves the record creators themselves because in-depth knowledge about the project usually fades quickly. But also future users who were not involved must be able to interpret the information.

Open-Source Viewers

During the BIM archiving exploration pilot, no budget was allocated for testing paid BIM viewers from market parties. Therefore, several widely used free or open-source viewers were used to get a good first impression of:

- What BIM/IFC information is.
- How it looks in viewers.
- What functionalities viewers generally offer.

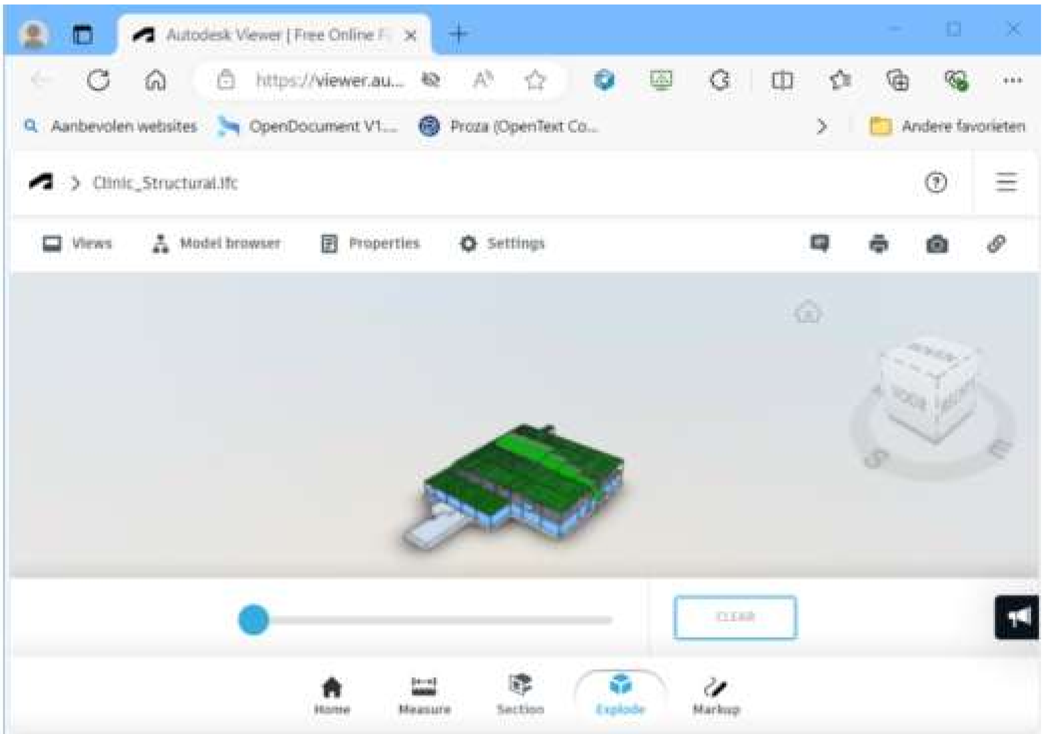

Figure 3. Free Autodesk Viewer.

During the project, the following viewers were tested in a test session:

- DDSCad Viewer (functional).
- Solibri Anywhere (blocked by security).
- Web viewer developed in 8 hours by a colleague of Ruud.

- Open IFC Viewer (does not work due to lack of GPU in the National Archives' virtual environments).
- BIMVision (does not work due to lack of GPU).

The free online test version of the Autodesk Viewer shows an IFC model and offers viewing options, see Figure 3. This viewer cannot be integrated into the National Archives' website because it is owned by Autodesk. However, functionalities such as zooming, rotating, selecting model parts or layers, viewing properties, and choosing types of views seem useful and are probably sufficient for an interested lay audience.

Costs

For budgeting a follow-up project, it is important to know what paid BIM viewers from market parties cost and how they compare qualitatively to widely used free or open-source viewers. Part of the pilot project note was to "explore the extent to which viewers such as, but not limited to, those from Future Insight are suitable for making such [BIM/ IFC] information accessible."

It is important here to distinguish between merely viewing and also editing and analyzing. BIM packages such as Clearly.BIM from Future Insight or, for example, Autodesk Revit or Graphisoft ArchiCAD cost thousands of euros per year and offer much more functionality than just viewing BIM. Think, in addition to display, of designing, cloud storage, analyzing, linking, sharing, and evaluating a BIM. A subscription to Clearly.BIM ("the smart online solution for easily viewing, querying, and sharing BIM models") costs, for example, €29,500 per year (https://www.futureinsight.nl/clearly-bim), and offers the following benefits:

1. An online solution for viewing, storing, and sharing a BIM.
2. Automated checking of a BIM against set requirements.
3. Adding different map layers to the Clearly.BIM environment.
4. Visualizing a BIM in a 3D Digital Twin.
5. Converting a BIM to, among others, CityGML.
6. Easily exporting a BIM in various formats.

For the archival sector, benefits 1 and 6 are particularly important: viewing, storing, sharing, and, if necessary, exporting the BIM/IFC model to other formats. This is also what most free viewers offer in terms of functionality.

You can check if a BIM/IFC model meets the minimum size requirements for a space. And also if doors and/or basic facilities are present. But this is not a priority for archives. Adding map layers to the Clearly.BIM environment is useful for showing BIM/IFC information from multiple transferred sources in context. However, it is unclear whether related models will be transferred. Similarly, the added value of visualizing a BIM in a 3D Digital Twin for the archival sector is unclear. There are few Digital Twins—especially within the national government—and even fewer in archived form at archives. We are also not aware of any connections between archives and Digital Twins at record creators, to visualize a transferred BIM/IFC model in those Digital Twins via such a connection. Converting a BIM/IFC model to CityGML is especially useful for integrating a BIM/IFC with and contextualizing a BIM/IFC in CityGML. This allows versions of a BIM/IFC design to be shown in the context of a neighborhood or city. This is only relevant for the archival sector if sufficient CityGML has been transferred. Or if there are connections from the archive to CityGML models. Viewed this way, archives would not use two-thirds of the functionalities of Clearly.BIM.

Due to the high costs of editing and analysis functionalities in BIM packages that are superfluous for the archival sector, and the presence of sufficiently similar functionalities in free viewers, the recommendation of this exploration is for the archival sector to prioritize the implementation of a free viewer.

Additionally, it is a good idea to promote collaboration between professional users of spatial data. This includes, among others, municipalities, the national government, water boards, and provinces. From such a collaborative framework, this software could become more widely available to a larger group of users.

Designated Communities

Two other considerations play a role here. First, the specific user group, or in ISO 14721 OAIS terms, the Designated Community, of the BIM/ IFC information that the National Archives offers as archives. Is this information intended for the general public without prior knowledge of and their own tools for BIM/ IFC? Or is it for a specialized expert audience with extensive prior knowledge and tools (in their "OAIS Knowledge Base")? Or a mix?

The National Archive in The Netherland has defined two Designated Communities for e-Depot certification, which take priority: the general public and the record creator who transfers government information to the service. The general public has little prior knowledge and tools, and the suppliers are not in-depth subject matter experts on all types of information and file formats they manage and transfer. However, they reasonably have a certain level of substantive, procedural, and contextual knowledge of what they transfer. From the perspective of these two Designated Communities, the preference should be to offer free viewers with sufficient functionality for viewing BIM/ IFC information. And thus not to offer BIM packages.

In the earlier chapter on BIM and public accessibility, a matter of principle arose due to the new Archives Act. When this law becomes active, a group of active users emerges. The examples mentioned were maintenance specialists and the fire brigade. It is currently difficult to place this group of users within the aforementioned Designated Communities. In follow-up research, these users must be included.

A second consideration concerns the technical infrastructure. The tests revealed that a video card (GPU) is a requirement for viewing BIM/ IFC information. Without a video card, as in an online IFC viewer, displaying and interacting with a model is very slow. Once the National Archives potentially shows many models to many visitors, and this is done from its own infrastructure, very powerful computers with video cards are needed to support this. Even then, the user needs a powerful computer to view the result in their browser. A look at the types of users also suggests that subject matter experts will likely already have powerful computers with BIM packages.

By choosing not only to view but also to edit and analyze in this infrastructure, the bar is set higher. This would significantly increase the costs of offering BIM/IFC information. While, as explained earlier, this is probably unnecessary given the sufficient functionalities in free viewers, the thinking about our Designated Community, and the presence of powerful computers among subject matter experts.

The intersection of free vs. paid, viewing vs. editing and analyzing, and general public vs. subject matter experts is an area for further research. For the short to medium term, choosing to offer free viewers in our infrastructure seems the best choice.

4. Accountability and Transparency in the Construction Process

The IFC standard does not include contextual and process information, which is crucial from an archival perspective. This information is essential for assigning authenticity, completeness, and integrity to BIM data. Such information can be recorded in the BIM Collaboration Format (BCF).

What is BCF?

BCF (BIM Collaboration Format) is an open data format, managed by BuildingSmart (like IFC), specifically designed for exchanging information and communication within BIM projects. The format enables the sharing of comments, issues, and changes in a BIM without sending the entire model files. BCF can be used as a BCF-XML file or as a structure for a BCF server.

How Does BCF Work?

BCF works by recording and sharing specific data about problems, changes, or comments in a BIM. This data includes:

- Title: A descriptive title.
- Text note: Detailed descriptions of problems, questions, or changes.
- Position information: Coordinates and display settings to mark the exact location in the model where a comment or issue is located. The current version of BCF does not yet include a provision for recording a translation of position information.
- Snapshots: A series of images that further illustrate the problem or comment.
- Comments: A series of text fields with metadata for user and date, not directly linked to other metadata such as status and priority.
- Status and priority: Information about the status (e.g., open, in progress, resolved) and priority of the issue or comment.
- Involved parties: Information about who is responsible for resolving the issue or addressing the comment.
- Involved IFC model and IFC objects: By recording the model to which the information in a dataset relates and the specific objects in that model, via Globally Unique Identifiers (GUIDs), the information in a BCF can be linked to data in an IFC.
- Documents: Within the BCF structure, documents can also be attached, including IFC snippets, small models representing specific changes that can be automatically processed by an application.
- No change history in BCF-XML: Only through the implementation of a BCF server can it be recorded who changed or can change which (meta)data. This information is not fixed in individual BCF-XML files.

How is BCF Used?

BCF is primarily used in BIM projects to improve communication and collaboration among stakeholders. Here are some ways BCF is used:

1. Problem Solving: When a problem is discovered in the BIM model, a BCF file can be created to document and share the problem with the team. This file can be uploaded to a shared environment accessible to all involved parties.
2. Coordination Meetings: During coordination meetings, BCF files can be used to discuss and track specific problems, ensuring a structured approach to problem-solving.
3. Quality Control: BCF can be used as part of quality control processes to ensure all problems are documented and followed up.
4. Integration with BIM Software: Many BIM software tools support BCF, allowing users to create, view, and edit BCF files directly from their modeling software, ensuring seamless integration into daily workflows.
5. Coordinate Changes: Proposed changes can be shared with modified IFC data in a BCF, commented on, and processed.

Who Uses BCF?

BCF is a relatively new standard for organizations in the built environment, with development starting in 2009. Unlike IFC, BCF is not yet an ISO standard. Many data platforms for building information use their own standards for communication and collaboration, but due to persistent demand, the most widely used applications and platforms can now use the BCF standard. However, when converting to and from BCF, certain aspects of communication specific to a particular application may still be lost.

Within organizations using BCF, most users are employees who benefit from direct access to the context of a specific communicated point to efficiently process changes. Where the focus of communication is more on securing accountability and financial considerations—most relevant for the National Archives—it is still common to make more generic agreements in separate documents.

An interesting development to monitor is the rise of Augmented and Virtual Reality (AR/VR), which could enable more substantive consultations with end users. Currently, reading and interpreting 2D drawings requires significant construction knowledge, limiting communication. Such AR/VR communication is already being captured in the BCF format in some applications.

For a complete overview of applications implementing a version of BCF, refer to the BuildingSmart Technical page: [Software Implementations - buildingSMART Technical].

Does BCF Fall Under the Archives Act?

One of the goals of an archive is to provide transparency. In the case of building information, you want insight into the construction process and the decisions made—when and by whom. For example, to reconstruct how a particular incident occurred. The BCF file can be seen as a metadata file containing such comments and decisions during the construction process. This can amount to hundreds of thousands of comments with varying content and purposes (see "How is it Used?"). It is likely that only a small part is relevant for long-term preservation. This is reinforced by the fact that not all layers of decision-making use BCF, and that employees in higher positions who do not work with BIM software often make important decisions.

Reconstructing a specific decision-making process from emails and other documents is a labor-intensive task. The use of standardized metadata, such as BCF, could be an important tool for ensuring the findability and reuse of BIM data. It could also promote correct communication within project teams and help prevent misunderstandings and inefficiencies. Miscommunication can lead to the loss of crucial information and inefficient working methods. However, as mentioned above, it is likely that the most relevant documentation of decision-making and processes for the archival sector is currently still found in separate documents. The BCF format thus has great potential to provide insight into the process and decision-making, but further research is still needed.

What Approach Could We Consider?

The above makes it clear that preserving all information recorded in BCF is probably not desirable. It quickly involves hundreds of thousands of comments from employees at different levels and with different purposes. This raises questions about selection and valuation, and at what level choices can and should be made. Obviously, due to the volume of comments, it is not possible to decide in detail which comments should or should not be permanently preserved. Since this will have to be done in broad terms, a capstone method could be considered. This would involve filtering based on the person who placed or was involved in a comment, and only preserving those from specific individuals. This could involve an entire topic, issue, or aspect of the building. It is also possible to choose to preserve topics where decision-making is involved in the process. But, for example, to omit topics that only concern quality controls.

However, a "standard" approach is still difficult to determine at this stage because BCF is not yet consistently used and applied. It is still a relatively new standard. A first step is therefore to increase awareness and knowledge of the applications and possibilities of BCF for every employee in the construction process, including those who do not come into contact with the software. Fragmentation of process information and communication in various forms such as emails and text documentation is not unusual. But the strong connection between IFC and BCF via GUIDs can greatly improve the accessibility and findability of this BIM process information, and further investigation of its use is therefore worthwhile.

Recommendations

In the previous chapter, the interfaces where BIM data overlaps with sustainable accessibility were discussed. Continuous development of knowledge is necessary and essential for the future of both the archival sector and the construction sector. The described challenges are not limited to the sustainable and accessible archiving of BIM data; they also play a role in multiple domains. This reinforces the necessity of explorations like this one.

In addressing the interfaces, the pilot members tried to answer the original research questions:

- To what extent does the Industry Foundation Classes (IFC) exchange standard lend itself to archiving and preserving BIM/As-Built information (sustainability)?
- To what extent are the viewers currently on the market suitable as viewers for making BIM data accessible (accessibility)?

The findings have now been discussed. However, many ifs and buts emerged from this. Therefore, we discuss below a number of recommendations for the follow-up process.

Collaboration with the Construction Sector: BIM and Archiving by Design

Despite the growing implementation of OpenBIM throughout the entire lifecycle of construction projects, the long-term preservation of this information is not well considered, according to experts. This is not surprising; it is quite standard that the needs of the executing parties differ from those of the archival sector. At the same time, this initial phase is precisely the point where the most influence can be exerted.

"Archiving by Design" means that during the design phase or during the expansion of construction software, the DUTO principles are taken into account. BIM software falls into two main groups:

- Factory design software, developed by companies like Autodesk, Solibri, or Tekla.
- BIM software developed with an open exchange standard: IFC from BuildingSMART like Qonic.

"Archiving by Design" can only be applied in combination with IFC. If archival and information professionals want archival principles to be part of the IFC standard, this must be organized within the community responsible for the creation of IFC: technical universities, governments, and (construction) companies, in other words, BuildingSMART.

"Archiving by Design" must become a fixed component in the (further) development of standards for open spatial data software such as IFC. Only in this way can Digital Spatial Data be sustainably accessible. This also applies to other open standards used in the construction sector. This means that "Archiving by Design" must be incorporated into the BuildingSMART community to ensure that this approach is standardly applied with every substantial adjustment to the standard. Fortunately, steps are already being taken in this direction: a project to link BuildingSMART and OpenBIM with the archive has already been initiated and proposed to the BuildingSMART development team.

In short, better collaboration is needed between the archival field and the construction sector. This goes both ways:

- The archival sector is needed to create awareness and shape the sustainable accessibility of BIM data before it is created.
- The construction sector is needed to interpret BIM data and make it understandable for archivists who often lack knowledge about it.

Practical Example: Guideline "Sustainable Digital Spatial Data"

In 2021, the Association of Dutch Municipalities (VNG), the Interprovincial Consultation (IPO), and the Union of Water Boards published the guideline "Sustainable Digital Spatial Data". This guideline addresses sustainable accessibility—by design—in the information chains of the Environmental Act for municipalities, provinces, water boards, and joint arrangements: environmental services, safety regions, and municipal health services. It contains:

"An extensive explanation of the background of sustainable accessibility and ends with a checklist of 34 requirements for this topic, organized into nine DUTO aspects. The requirements are divided into what is legally required, what must be done by the implementation date of the Environmental Act, and what can contribute to sustainable accessibility."

Creating Support Within the Archival Sector

To make broadly supported decisions about the issues mentioned in this report, collaboration within the archival sector is logically also important, in addition to collaboration with the construction sector.

In follow-up research, we can examine how Dutch archival institutions deal with this matter. How prepared are organizations for managing BIM data? There are likely significant differences in the level of knowledge between (local) archives. We can also look at examples from abroad, where, for example, commercial software is used to provide access to IFC models.

Examples of organizations to seek collaboration with include:

- Municipal archives (individual and VNG).
- Regional archives.
- Water boards.
- International (the e-Ark Foundation and the DILCIS Board).

Heritage and data/scientific organizations:

- The New Institute.
- Cultural Heritage Agency of the Netherlands (RCE).
- Geonovum.
- TNO.

Steering Data Quality and Richness

In the sections on the flexibility of the IFC standard and viewers, it emerges that the IFC standard offers a great deal of freedom. To improve sustainable accessibility, we must focus on improving data quality within IFC files. Clear work instructions on how to fill in these standards are still lacking. The pilot also revealed that it is not sufficient to prescribe IFC alone as an archiving standard for BIM data. It is important to establish requirements in which supporting documentation, specifications, and files are requested.

We can draw on projects that have already addressed this. An important example at the national level is the "Start Architecture: Sustainable Reuse of Spatial Data". Internationally, DURAARK has been focusing on improving data quality and metadata in the spatial sector and supporting IFC as an archiving format since 2014.

BuildingSMART also seems aware of these challenges and has developed several supporting standards and specifications that are important for contextualizing and interpreting IFC. The main examples are the OTL and ILS. Recently, the Information Delivery Specification (IDS) was accepted as a new standard, in which BuildingSMART emphasizes that construction companies must stop inventing their own properties that already exist in the IFC standard. This can also be linked to the Building Smart Data Dictionary (bSDD). We have identified BCF as a standard format for communication. Ideally, at the end of the project, a document will be produced describing the process, purpose, and decision-making.

Furthermore, it is important to steer towards the use of newer versions of the IFC standard.

Advisory organizations often lag behind the facts. For example, the Standardization Forum currently still prescribes version 2x3 TC1, but IFC4X3_ADD2 (ISO 16739-1:2024) is the current version. This is not due to unwillingness or inability on the part of these organizations but is a consequence of the rapid developments in the standard.

For the archival field, the task is therefore to specifically define what information is needed and in what forms this information should be received. This is described in the next paragraph.

In follow-up research, the project group wants attention to be paid to the Information Delivery Specification (IDS). Due to time constraints, this topic could not be explored in depth during this pilot. IDS is a crucial component within the Building Information Modeling (BIM) methodology, specifically aimed at structuring information exchange. IDS defines the specifications for the information that must be delivered at certain times. This ensures that all involved parties have clear and consistent requirements for what information must be available at what time, allowing projects to be executed more efficiently and with fewer errors. The process around it is described in an IDM (Information Delivery Manual). IDS offers us all kinds of new possibilities.

Accepting a Degree of Data Loss

Choosing IFC as an archiving standard requires creating comfort with a certain degree of data loss, which goes against the current vision in the Dutch archival sector. Preserving digital archives in their original state is the norm. Archiving in an exchange standard is also not yet done in the Netherlands. Although it is possible to choose to deliver in both native formats (with full data richness) and an IFC manifestation (loss of some functionalities), this brings other challenges due to the strong use of proprietary software and the size of such files. A good consideration of including different manifestations and the costs involved (not only financial) must be made. It should be a calculated decision that is widely supported. The IFC standard is also evolving in this regard, and new versions are visually better and data quality is improving.

In this way, it also plays into a question that arises in multiple domains. Specialists often work with commercial state-of-the-art software packages, which is easiest for them to use off-the-shelf with a software license agreement. Open-source software usually requires more effort in setting up work processes, and how support from the maker of this software is arranged is still uncertain. By choosing IFC, the archival sector accepts that a certain degree of data loss will occur.

User Research

Finally, it is important for the archival sector to gain more specific insight into who the "users" will be for this domain and what they need in terms of access to this information. There is a public function, but also an accountability function that needs to be shaped, and there are different levels of access to be realized. For example, due to security and privacy considerations.

The wishes of the "customer/user" can vary greatly. For example, reuse may be the goal, but there are also organizations (such as police and fire brigade) that may need access to BIM data for safety reasons, possibly with a high degree of urgency. Additionally, the archived information must be able to serve as accountability and transparency about the construction process and decision-making, especially in the event of an incident. And it must be able to serve as a memory, for example, during a renovation to reconstruct what happened.

Earlier in the advisory report, the designated community "general public" was mentioned. It is good if follow-up research focuses on this target group.

A rough estimate of interested parties includes:

- Residents who want to know everything about their house.
- Researchers: What research questions do they bring?
- Design firms.
- Architecture firms.
- Provincies.
- Rijkswaterstaat (Dutch Directorate-General for Public Works and Water Management).
- Municipalities.
- Kadaster (Dutch Land Registry, possibly in an exemplary function).
- Government Real Estate Agency.

For this, use cases can be set up, where research is conducted into what levels of access are needed for the different cases, and how we can link the geometric data and context.

This also presents an opportunity to give direction to the course of the KVAN with "the archive as a utility". By reasoning based on standards and from the user's perspective, regardless of the angle from which the information was formed. With the utopia that we can make BIM data easily accessible to everyone, regardless of technical knowledge, using a viewer.

Conclusion

This advisory report marks a starting point, a first exploration of the sustainable and accessible archiving of BIM data in the IFC format. There is still a long way to go to actually implement this. By reporting findings and giving recommendations, the project team hopes to give the topic a significant boost, especially since many of the challenges also play a role in other domains. For example, during the preparation for the transfer of geodata from offshore wind farms, similar challenges arose regarding the sustainable accessibility of raw data, the use of viewers, and the role of the designated community.

While we, as project partners, continue to work on this topic within our organizations, we will also communicate externally. The pilot has shown that there is interest in this theme. Among archivists, knowledge needs to be built, and this is best done in collaboration. A guideline on the sustainable accessibility of BIM could be a nice follow-up.

Finally, it is important that this knowledge also receives more attention within the construction sector. Important decisions are made in the sector regarding the development of construction software and the implementation of standards. From the perspective of "Archiving by Design", it is important that archival and information professionals in this sector come to the fore. After all, it is in the interest of the construction sector itself that BIM is managed in a sustainably accessible manner, but awareness of this still needs to grow. We must prepare ourselves for the sustainable and accessible management of digital information. How else can we guarantee the information provision and maintenance of new and existing construction projects?

Appendix: Requirements Testing Session

Open-Source Viewer Requirements for IFC Files

Within the BIM exploration, we are working out various scenarios. One of the scenarios we are looking at, which corresponds to current practice, is recommending specific open-source viewers to users to display information objects.

Below is a roughly sketched list of requirements that we could impose on such a viewer. These requirements were used during the test session on June 4, 2024.

We prefer the threshold for accessing information objects to be as low as possible, so in another scenario, we are working out what implementations could take place from the archival field to, for example, provide access within a website or portal.

IFC Viewer Requirements

Viewer is free to download/use (no licenses).

Viewer preferably has an active user group and receives (regular) updates.

Viewer provider does not request personal data for use.

When investigating possible free viewers/editors for the IFC format, we encountered that in almost all cases, an email address must be provided before the software can be downloaded. Providing an email address can be experienced as a threshold by our broad group of users. I would like to discuss what everyone thinks of this; it may be a non-issue or insurmountable.

Viewer makes data and metadata insightful

An obvious requirement that needs to be further refined and specified regarding BIM data and/ or the IFC format. The viewer at least provides insight into how the data was created, by whom the data was created, and how it was used.

Challenges:

- IFC seems to have heterogeneous and inconsistent metadata schemas.
- Displaying differences between "as-planned," "as-built," and "as-is"?

Possible important functionalities that support insightfulness:

- Displaying measurements between specific points.
- Zoom/pan/rotate.
- Isolating and/or hiding objects.
- Different views for different users: 3D, floor plan, etc.

- [Please add suggestions from experts].

Viewer poses no security threat

Low-hanging fruit, but possibly still important to mention.

Essential characteristics are displayed authentically

In digital preservation, we regularly look at the essential characteristics (significant properties) of a group of information objects, often at the level of information type. As the name suggests, it concerns (behavioral) elements of the information object that are of essential importance when rendering the object. On the Github of the National Archives and Records Administration (NARA), you can find an example of the essential characteristics of geodata. We could investigate and record such a list for BIM data, or more specifically for IFC models, if this fits within the scope of the exploration. And next to this, determine whether the viewer offers the desired display or functionality.

Viewer allows protecting partially public material

We have discussed that it may be the case that a BIM model is either entirely or partially restricted. Fully closed objects are not offered as downloads, but is this for partially open, partially closed BIM models something that should also be taken into account when choosing a viewer? Or is this regulated at an earlier stage, and does a viewer have little influence on this?

Viewer supports different versions of the IFC format

The viewer preferably supports as many versions/ schemas of the IFC model as possible, and in particular IFC 4X3.

Questions about this requirement:

- Do you think we also need viewers for other schemas, such as IFC4.1, IFC4.2, etc.?
- Should we be more specific than "IFC 4X3"? I also see "IFC4X3_RC1" and "IFC4X3_RC4" mentioned, for example.

Viewer allows validation

This is more of a question than a requirement; we have already discussed the importance of validation and its importance for BIM data. Is it then also necessary that an open-source viewer has this functionality? Or is this rather something that requires a separate validator?

A possible "anti-requirement": editing the model is not needed?

In our previous meeting, there was a distinction between IFC viewers and editors. Is the assumption correct that editing the models is not necessary when providing access? Or are there user groups that do need this?

The Open IFC Viewer from the Open Design Alliance (ODA) seems a promising possibility. Especially due to its affiliation with BuildingSMART. This has also come up several times in our conversations. Therefore, I would like to hear experiences and opinions about this viewer again and whether this could be a good option for the viewer test in this scenario.